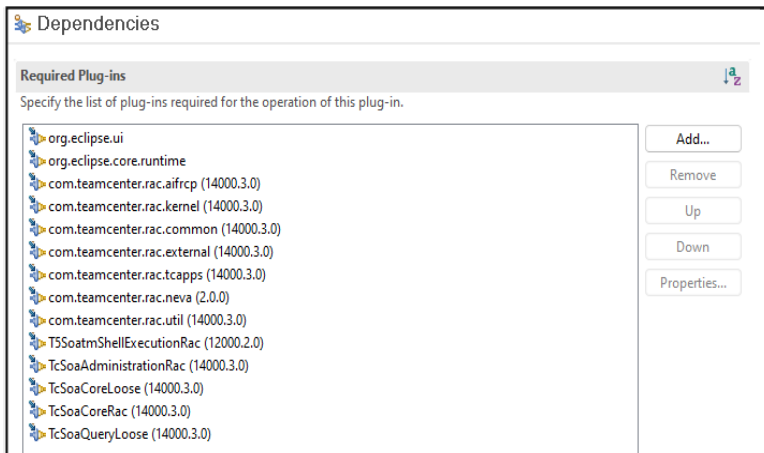


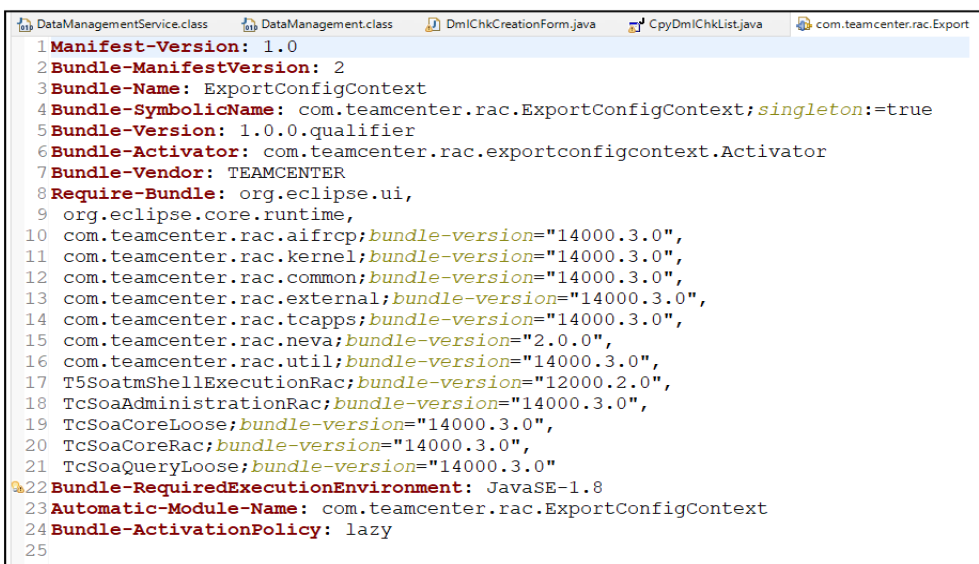
RAC Development manual.

1.) List all the major TC dependencies required for Teamcenter RAC development?



2.) What is the use of Meta INF file?

Meta INF file list down all the available TC dependencies which are used in Project.



3.) What is the use of Plugin.xml in RAC Project?

Plugin.xml file is used to declare and define menu commands there uses and location in Teamcenter menu bar.

Plugin.xml file serves as a connecting link between front and backend Java code.

each click on menu command executes a Java code file with is linked to that command in plugin.xml file.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <?eclipse version="3.4"?>
3 <plugin>
4
5     <extension
6         point="org.eclipse.ui.commands">
7         <category
8             id="com.teamcenter.rac.ExportConfigContext.commands.category"
9             name="Sample Category">
10         </category>
11         <command
12             categoryId="com.teamcenter.rac.ExportConfigContext.commands.category"
13             name="Download Applicability matrix"
14             id="com.teamcenter.rac.ExportConfigContext.commands.sampleCommand">
15         </command>
16         <command
17             categoryId="com.teamcenter.rac.ExportConfigContext.commands.category"
18             name="Compare Vehicle/Variants Report"
19             id="com.teamcenter.rac.ExportConfigContext.commands.sampleCommand2">
20         </command>
21     </extension>
22     <extension
23         point="org.eclipse.ui.handlers">
24         <handler
25             class="com.teamcenter.rac.exportconfigcontext.handlers.SampleHandler"
26             commandId="com.teamcenter.rac.ExportConfigContext.commands.sampleCommand">
27         </handler>
28         <handler
29             class="com.teamcenter.rac.exportconfigcontext.handlers.SampleHandler2"
30             commandId="com.teamcenter.rac.ExportConfigContext.commands.sampleCommand2">
31         </handler>
32     </extension>
33 </plugin>
34 </extension>
35

```

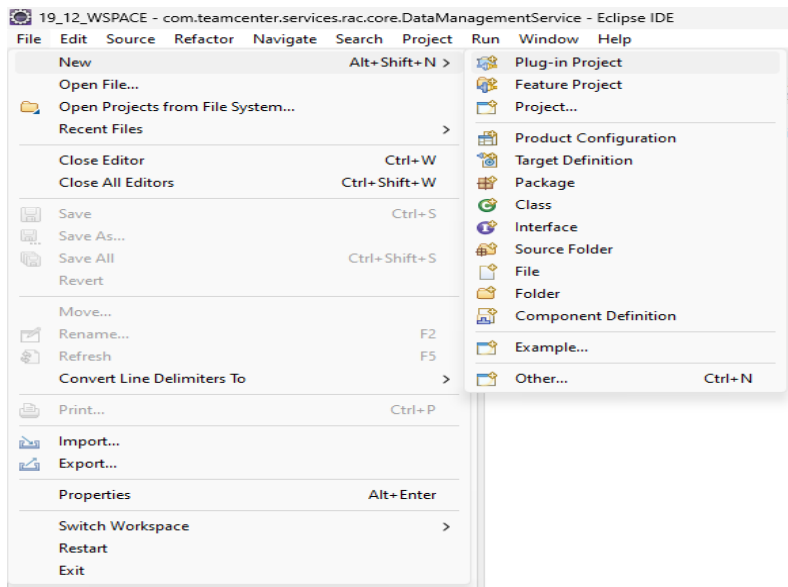
```

19         id="com.teamcenter.rac.ExportConfigContext.commands.sampleCommand2">
20     </command>
21 </extension>
22 <extension
23     point="org.eclipse.ui.handlers">
24     <handler
25         class="com.teamcenter.rac.exportconfigcontext.handlers.SampleHandler"
26         commandId="com.teamcenter.rac.ExportConfigContext.commands.sampleCommand">
27     </handler>
28     <handler
29         class="com.teamcenter.rac.exportconfigcontext.handlers.SampleHandler2"
30         commandId="com.teamcenter.rac.ExportConfigContext.commands.sampleCommand2">
31     </handler>
32 </extension>
33 </extension>
34 <extension
35     point="org.eclipse.ui.menus">
36     <menuContribution locationURI="menu:tools.Product Structure Action?after=Supplier Data Sharing">
37         <menu label="Product Structure Action" id="tools.Product Structure Classes">
38             <menu id="com.teamcenter.rac.ExportConfigContext.menus.sampleMenu"
39                 label="Product Configurator Reports" icon="icons/sample.png"
40                 mnemonic="M">
41                 <command commandId="com.teamcenter.rac.ExportConfigContext.commands.sampleCommand"
42                     id="com.teamcenter.rac.ExportConfigContext.menus.sampleCommand"
43                     mnemonic="S">
44                 </command>
45                 <command commandId="com.teamcenter.rac.ExportConfigContext.commands.sampleCommand2"
46                     id="com.teamcenter.rac.ExportConfigContext.menus.sampleCommand"
47                     mnemonic="S">
48                 </command>
49             </menu>
50         </menu>
51     </menuContribution>
52 </extension>
53

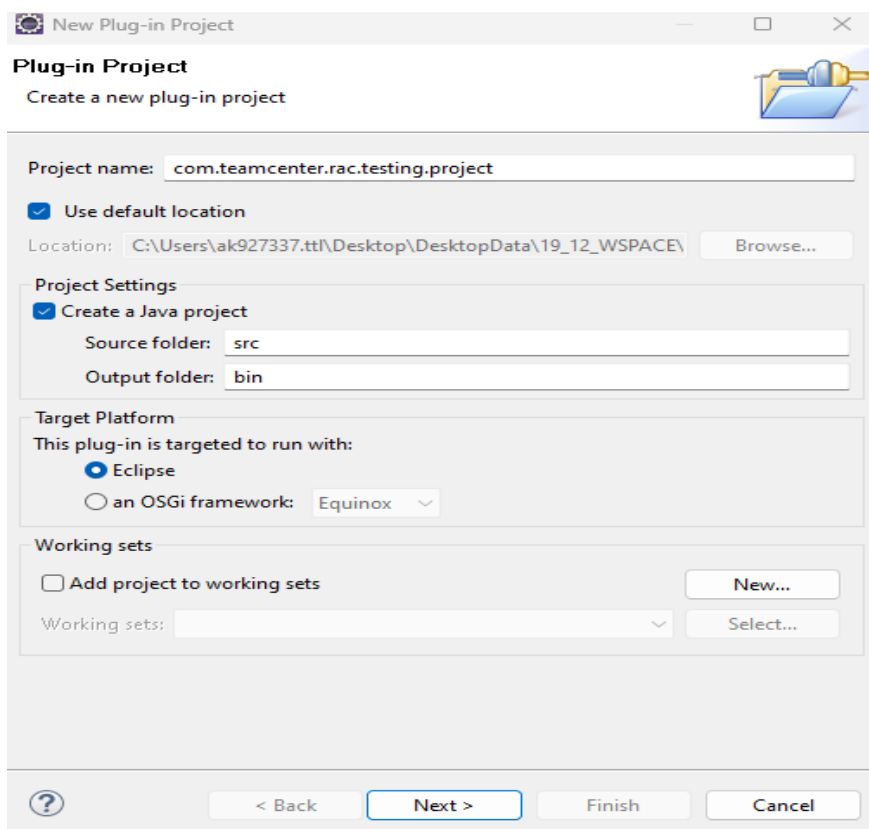
```

4.) How to create a RAC Project?

Step 1.) Open Eclipse IDE File -> New > Plugin Project.



Step 2.) Name the project for example like (com.rac.teamcenter.rac.project.testing)



Step 3.) Select Java Version, Select Generate and Activator check box as shown below.

New Plug-in Project

Content
Enter the data required to generate the plug-in.

Properties

ID:

Version:

Name:

Vendor:

Execution environment: [Environments...](#)

Options

☒ Generate an activator
Activator:

☒ This plug-in will make contributions to the UI

☐ Enable API analysis

Rich Client Application
Create a rich client application? ☐ Yes ☒ No

[?](#) [< Back](#) [Next >](#) [Finish](#) [Cancel](#)

Step 4.) Click on check box as shown below and select “Hello world command”

New Plug-in Project

Templates
Select a template to generate a fully-functional plug-in.

☒ Create a plug-in using a template

Available Plug-in Templates:

- Custom plug-in wizard
- Editor contribution for XML files
- Hello, World Command**
- Incremental project builder
- Menu contribution using 4.x API
- Multi-page editor
- ODA Data Source Designer
- ODA Data Source Runtime Driver
- Plug-in with a DTP connection profile
- Plug-in with a DTP driver template
- Plug-in with a DTP JDBC Connection Prof
- Property page
- Sample help content
- Textual editor, relying on Generic Editor
- Toolbar contribution using e4 API
- View contribution using 3.x API
- View contribution using 4.x API
- View using browser technology

This wizard creates a standard plug-in directory structure and adds the following:

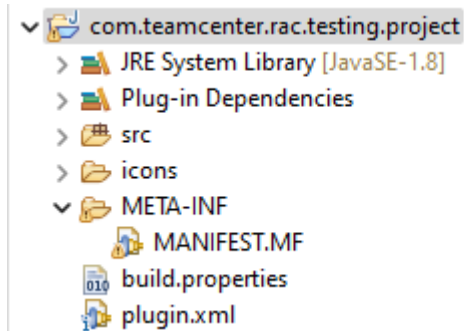
- Command contribution.** This template creates a simple command contribution that adds **Sample Menu** to the menu bar and a button to the toolbar. Both the menu item in the new menu and the button invoke the same **Sample Action**. Its role is to open a simple message dialog with a message of your choice.

Extensions Used

- org.eclipse.ui.commands
- org.eclipse.ui.handlers
- org.eclipse.ui.bindings
- org.eclipse.ui.menus

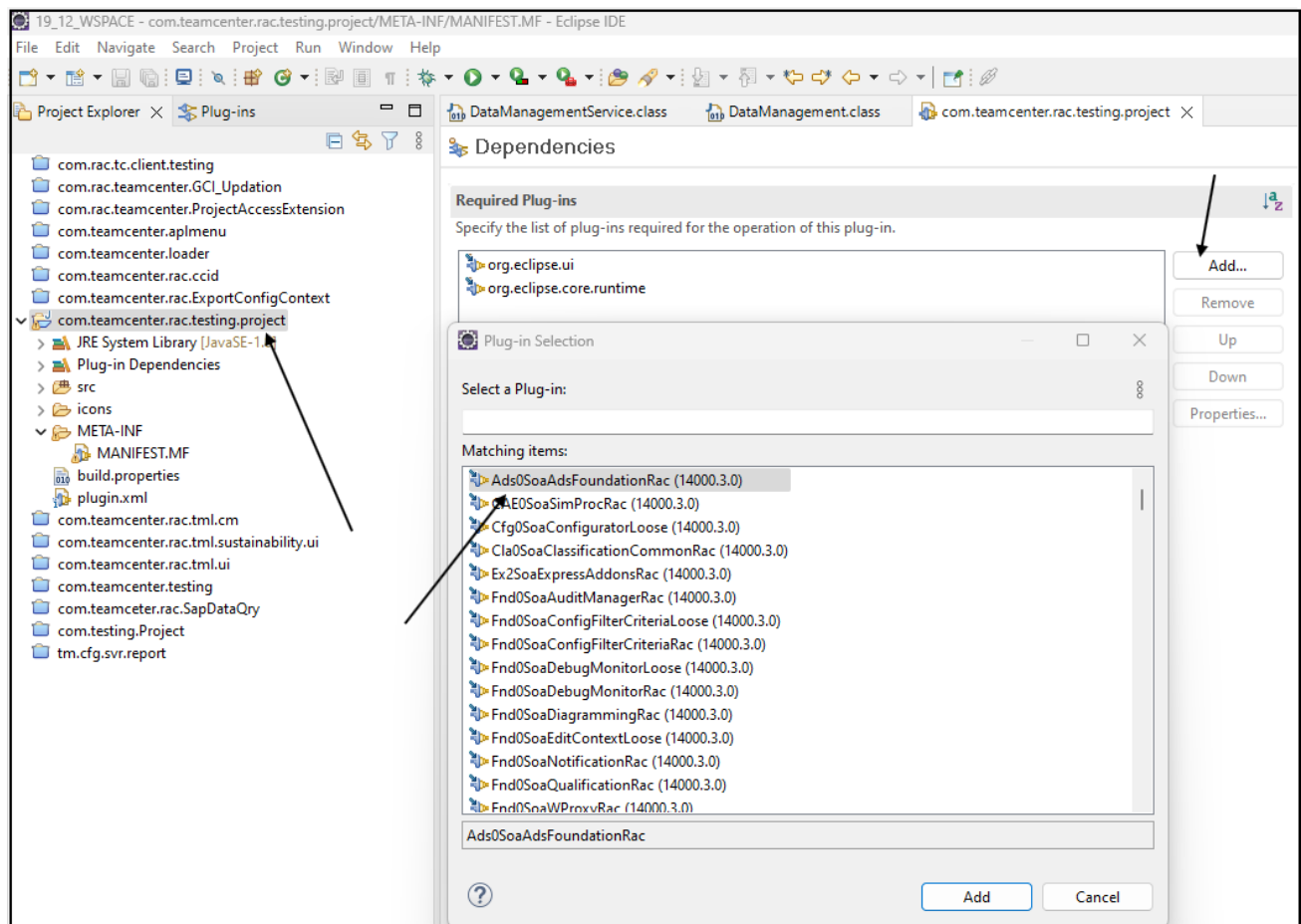
[?](#) [< Back](#) [Next >](#) [Finish](#) [Cancel](#)

Step 5.) Click on next and finish the set up. New project will be created and added in explorer view of Eclipse.



5.) How to add Dependency to a new RAC Project?

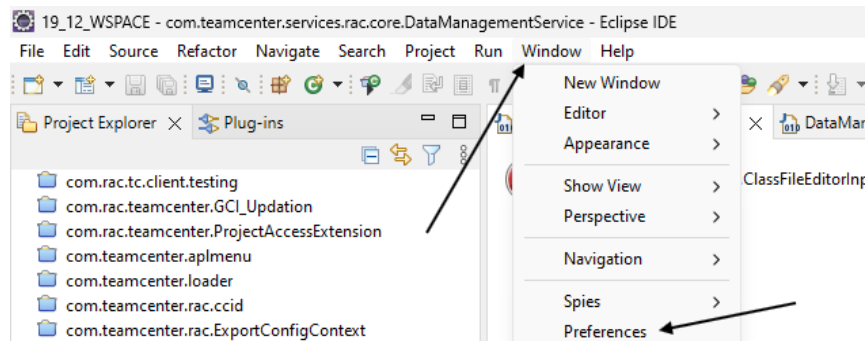
To Add dependencies, go to dependency tab in Project and dependencies to Project.



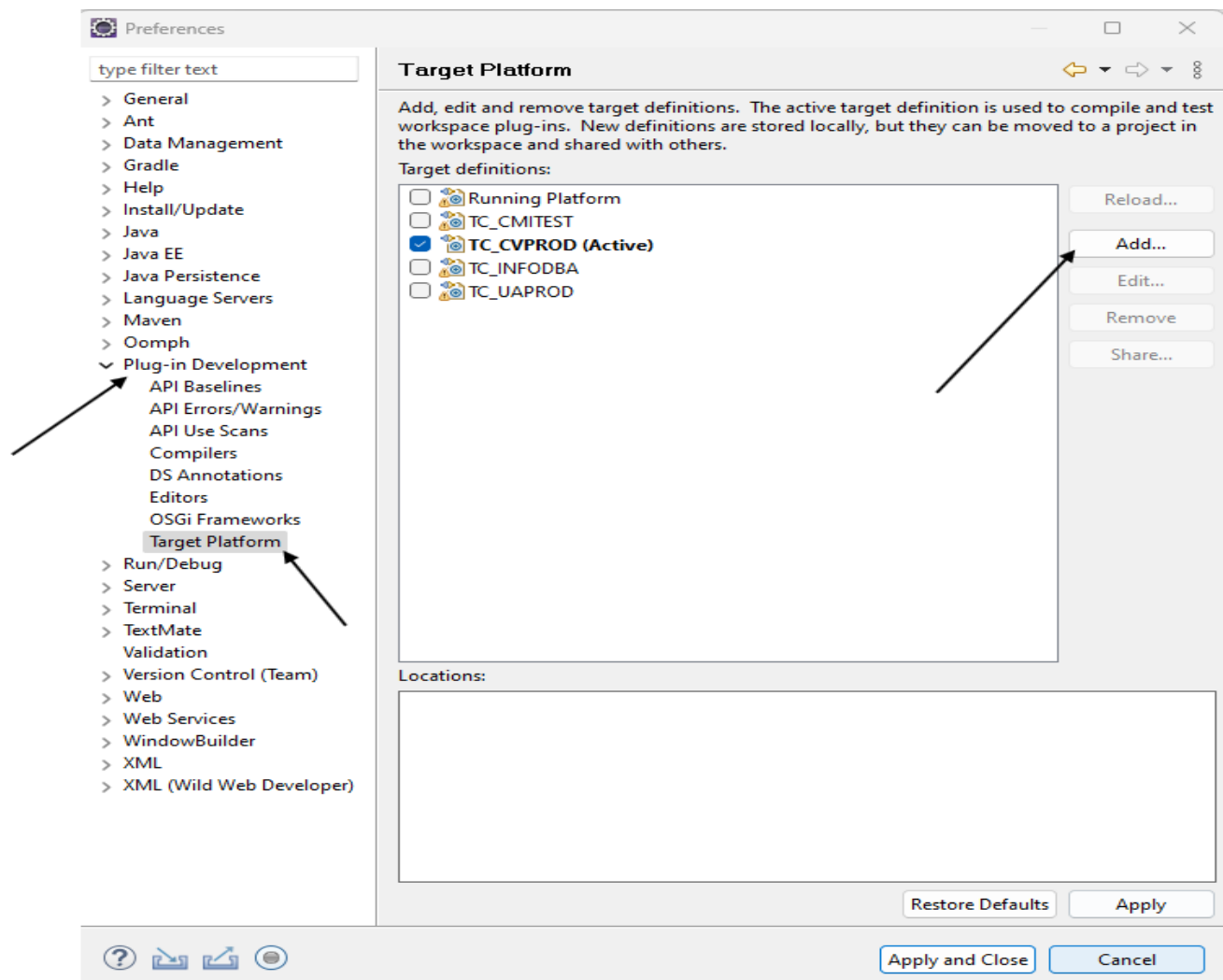
6.) How to add Teamcenter dependencies to Eclipse IDE?

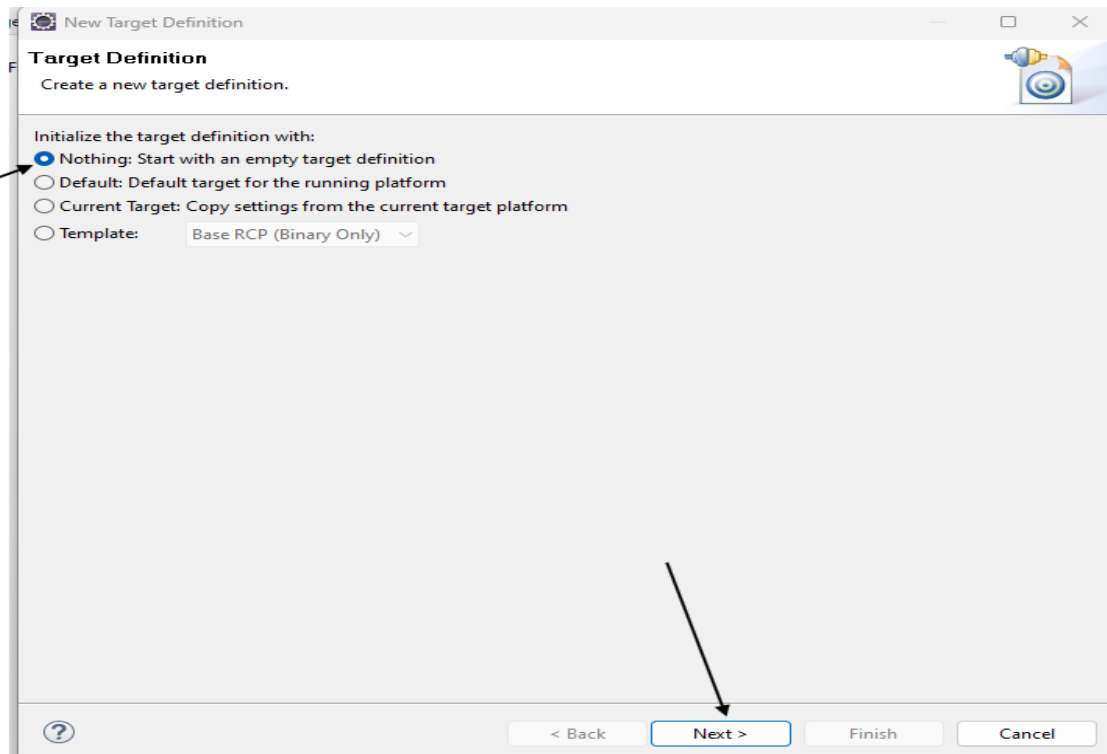
Step 1.) Add Plugin Folder Path of Teamcenter

Window -> Preferences

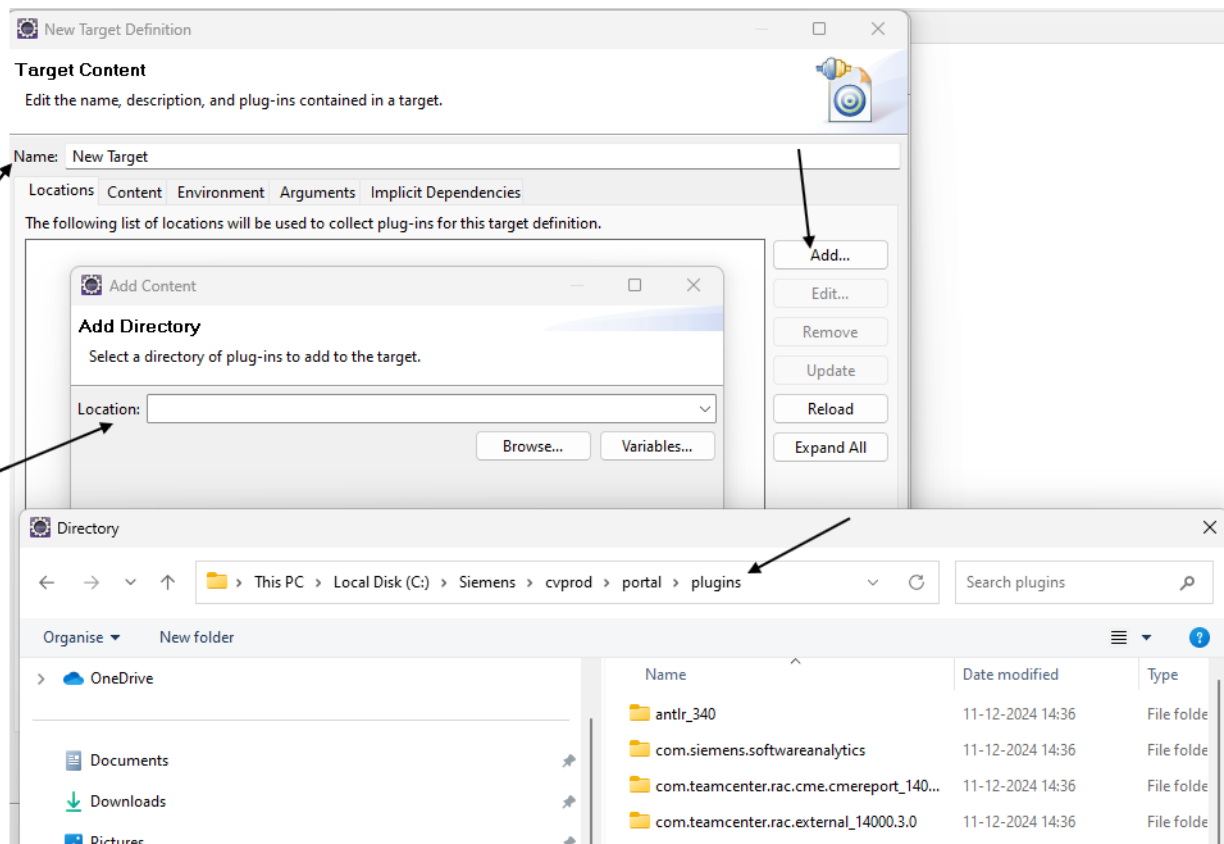


Step 2.) Go to Plugin development -> Target Platform -> Click on “Add”

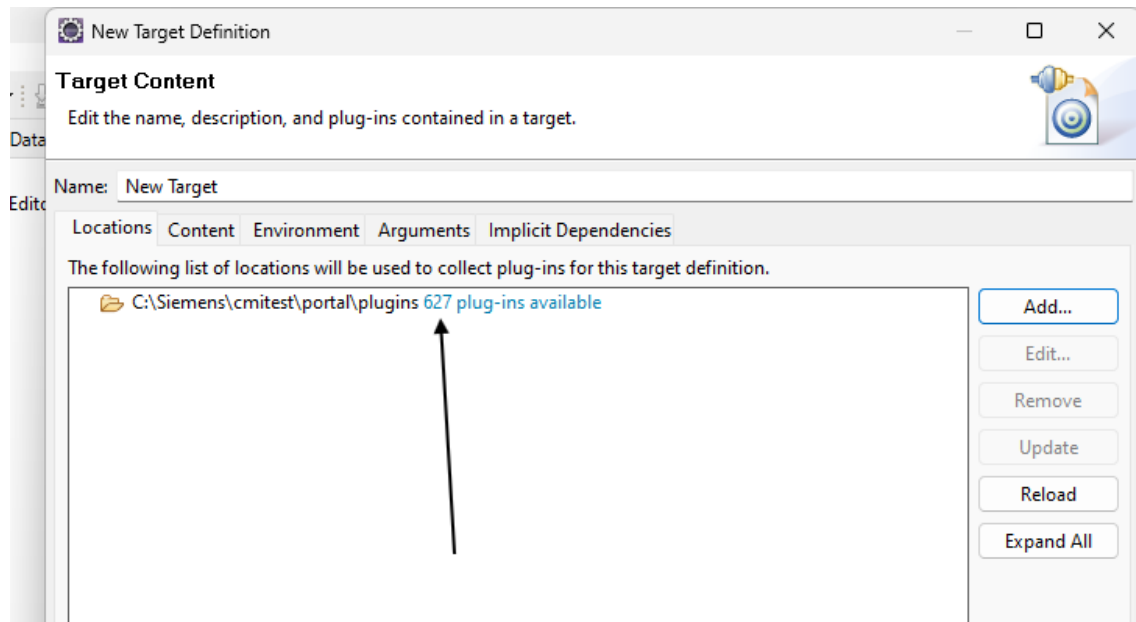




Step 3.) Name the target platform -> Add Location -> Path of plugin folder of teamcenter



Step 4.) Once Libraries are considered in eclipse available dependencies will be displayed -> finish the set up.



7.) What are major build components of application window available in Java?

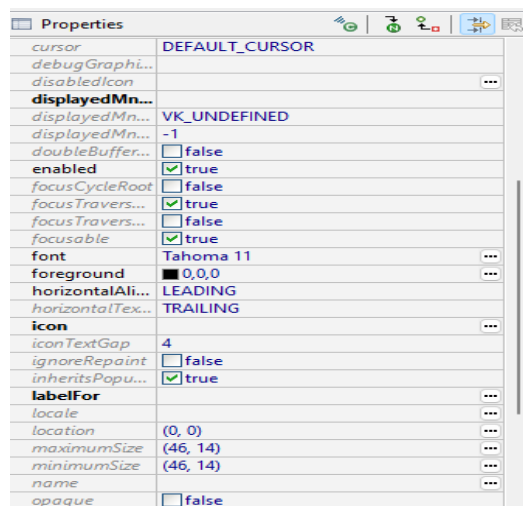
1.) JLabel, JButton, JTable, JTextField, JComboBox, JCheckBox, JRadioButton etc .

8.) What are the major layouts of application window available in Java?

1.) JPanel, JScrollPane, JSplitPane, JToolBar

9.) How to Edit Component in application window?

To Edit components, go to properties tab and edit available properties.



10.) Name some major classes used in TC RAC development?

TCComponent, TCSession, TCService, TCComponentQuery, TCComponentQueryType.

11.) Explain TCSession class in detail?

TCSession class is used to get the current login session of user. With object of this class, we can know current is logged in user id, current group and current role, hostname etc.

12.) Explain TCComponent class in details?

TCComponent class is parent class of all Teamcenter objects. Object of this class can be casted to Item, Item revision, Dataset etc.

13.) What is the use of sample handler.java and how it is used in Plugin.xml?

SampleHandler.java is the main code which is called once a command is activated in UI of teamcenter. SampleHandler.java and Teamcenter UI command are linked with each other in Plugin.xml

14.) What are ActionListener? Explain in detail?

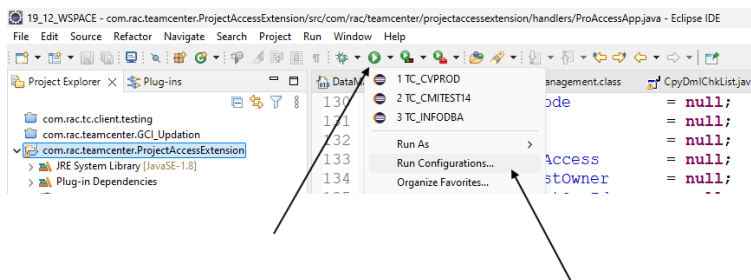
ActionListener is an interface found in the **java.awt.event** package that is used to handle action events. These events typically occur when a user interacts with a graphical user interface (GUI).

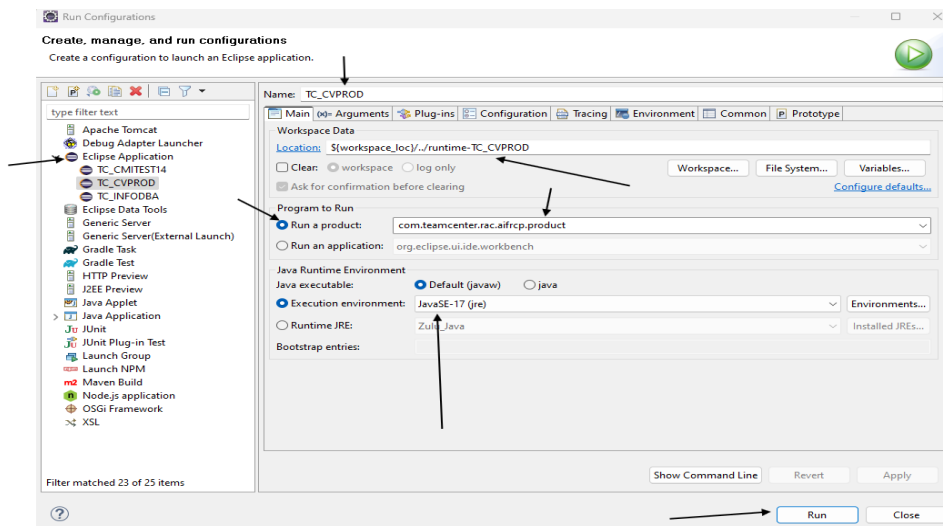
Example

- 1.) Clicking on mouse, pressing keys, moving cursor etc.

Listeners are added by right clicking on component -> Add ActionListener command.

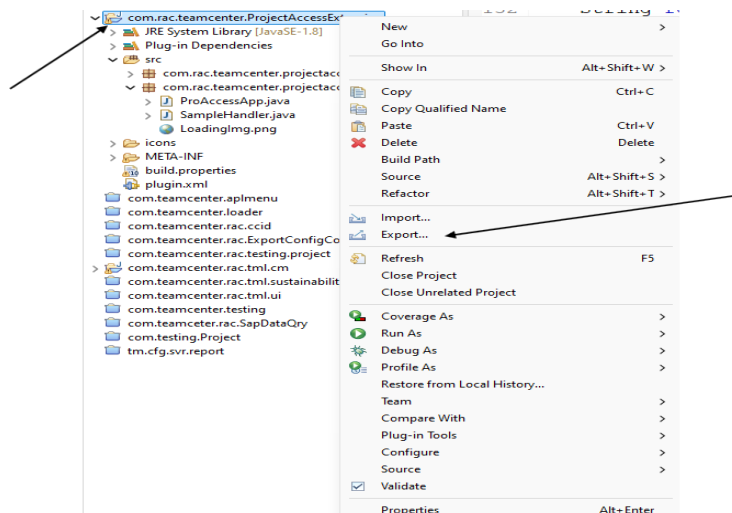
15.) How to run the Project in Eclipse IDE?



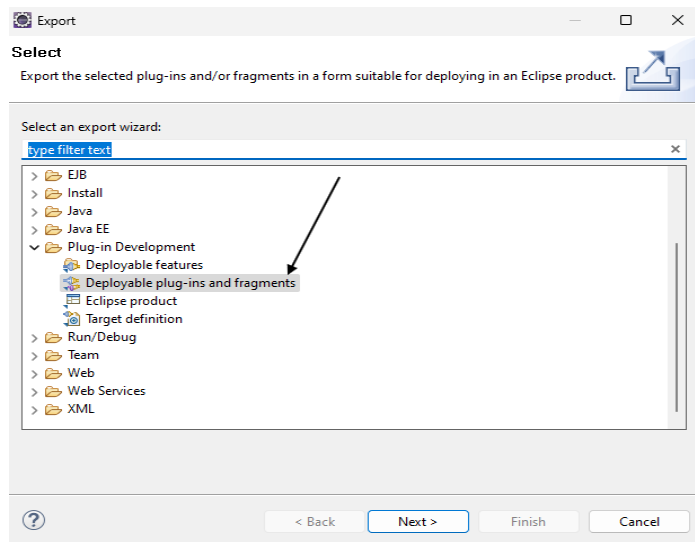


16.) How to build TC RAC project and generate Jar files?

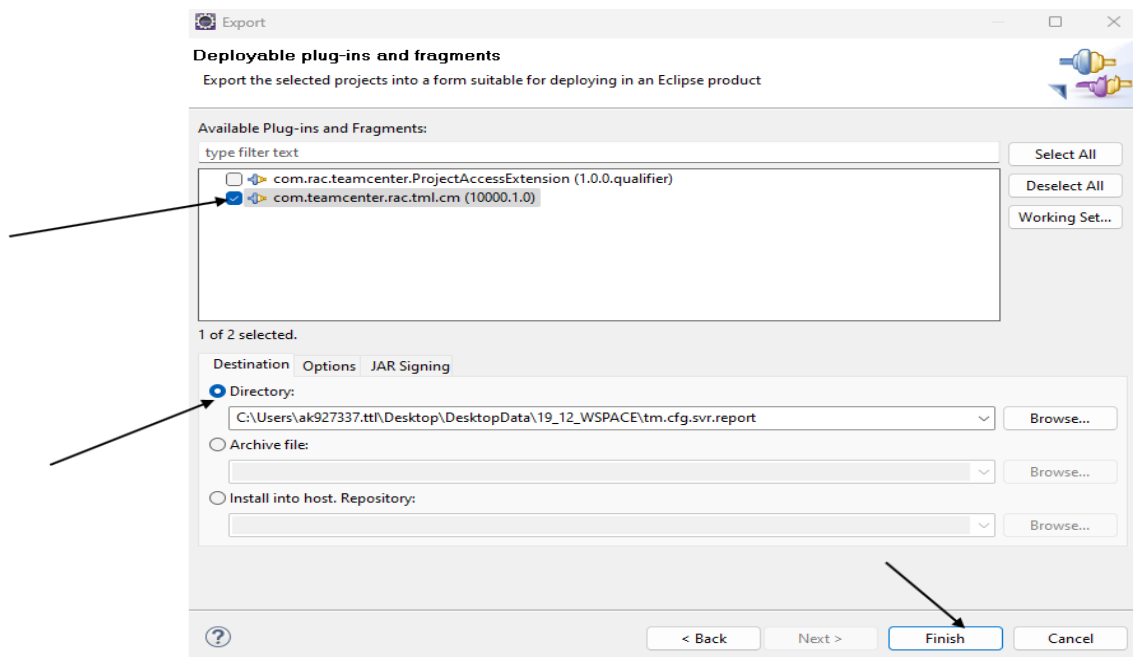
Step 1.) Right click on project



Step 2.) Click on Deployable Plug-ins and fragments



Step 3.) Select the project -> select a directory to generate output files -> finish

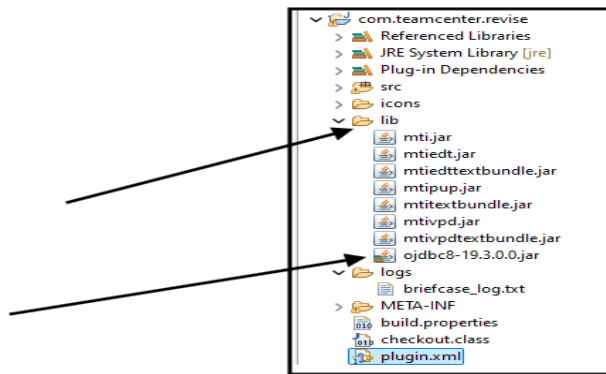


17.) How to deploy generated Jar file in Teamcenter?

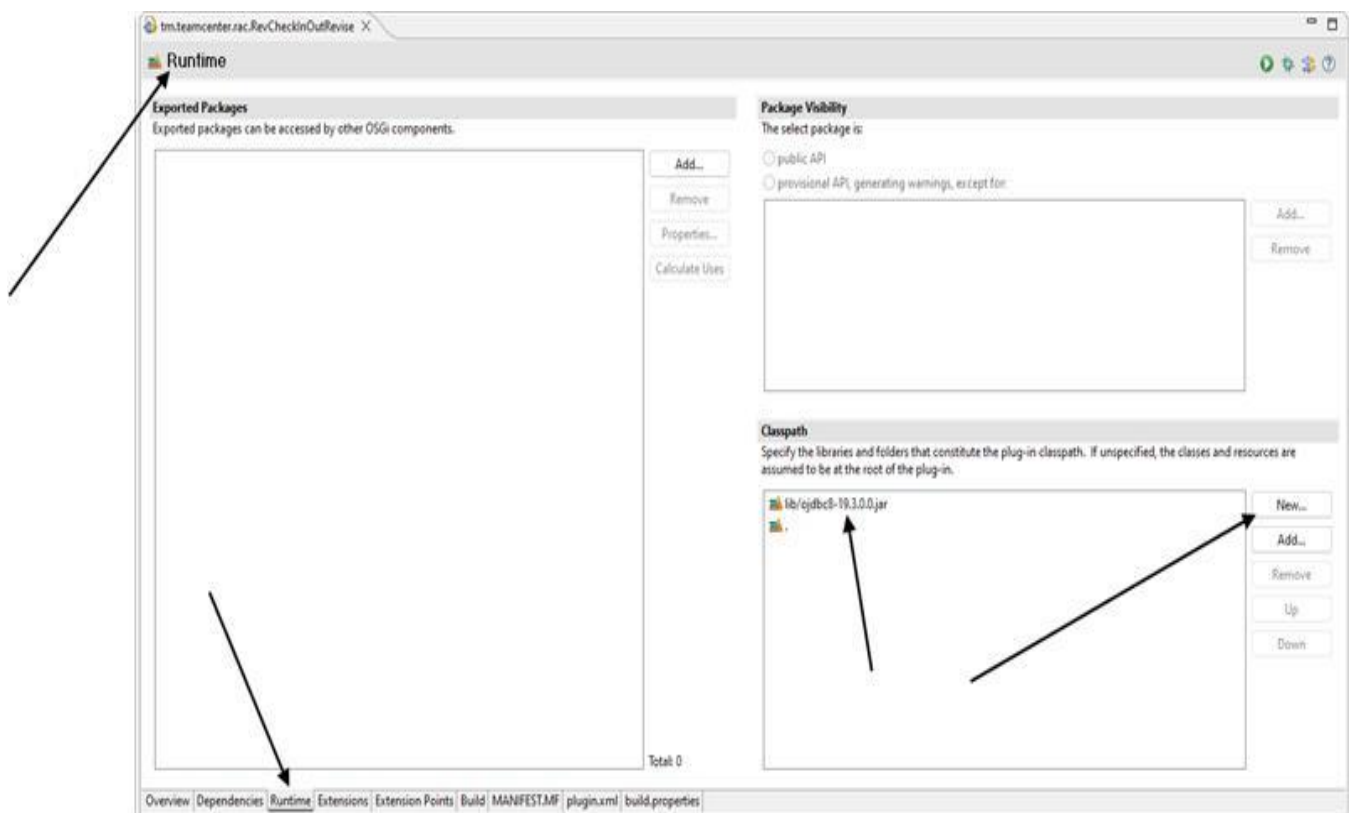
To deploy Jar files, copy Jar file in Plugin folder -> Run registry through ("C:\Siemens\cvprod\portal\registry") -> Restart teamcenter .

18.) How to make JDBC connection with Oracle database?

Step 1.) Create lib folder and add required external Jar files in it.



Step 2.) Add JDBC jar with Runtime configurations in Project.



Step 3.) Add connection function with required Server IP address, Logging and password.

```
public Connection getDBConnection()
{
```

```
    String ip_oracle="172.22.97.13";
    String port_oracle="1521";
    String SID="PFTEST";
```

```
    DBURL="jdbc:oracle:thin:@"+ip_oracle+": "+port_oracle+": "+SID;
    String DBUSER="xclusterdb";
    String DBPASS="xclusterdb23";
```

```

    Connection conn = null;
    try {
        Class.forName(DBDRIVER);
    } catch (ClassNotFoundException e) {
        // TODO Auto-generated catch block
        e.printStackTrace();
    }
    System.out.println("BEFORE DB CONNECTION");
    try {
        conn = DriverManager.getConnection(DBURL, DBUSER, DBPASS);
    } catch (SQLException e) {
        // TODO Auto-generated catch block
        e.printStackTrace();
    }
    System.out.println("DB Connected");

    return conn;
}
public static void closeConn(Connection conn)
{
    try{
        if(conn != null){
            conn.close();
            System.out.println("Connection close");
        }
    }
    catch(Exception e){
        System.out.println("|||||||ERROR|||||||");
        e.printStackTrace();
    }
}
}

```

19.) How to open a POP-Message on the start of teamcenter?

To open a message or dialog in starting of Teamcenter use startup dependency

```

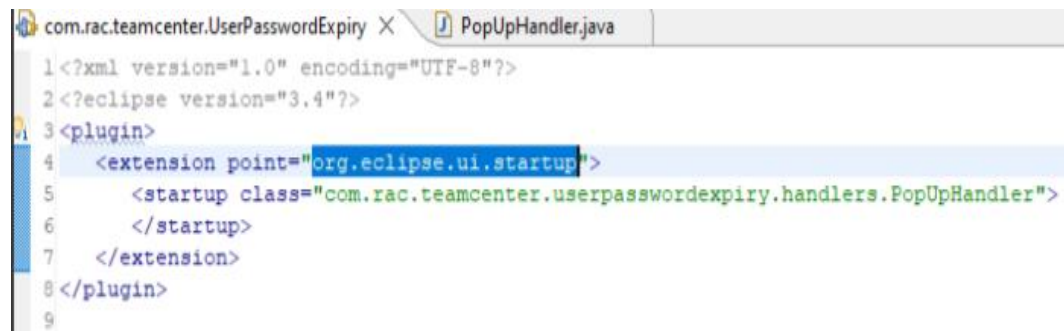
<?xml version="1.0" encoding="UTF-8"?>
<?eclipse version="3.4"?>
<plugin>
<extension point="org.eclipse.ui.startup">
<startup class="com.rac.teamcenter.userpasswordexpiry.handlers.PopUpHandler">

```

</startup>

</extension>

</plugin>



```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <?eclipse version="3.4"?>
3 <plugin>
4   <extension point="org.eclipse.ui.startup">
5     <startup class="com.rac.teamcenter.userpasswordexpiry.handlers.PopUpHandler">
6     </startup>
7   </extension>
8 </plugin>
9
```